

SONNEBERG NEUFANG, Germany

WMO: 105580

Lat: 50.383N

Lon: 11.183E

Elev: 630

StdP: 93.98

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-14.3	-11.4	-18.5	0.8	-11.3	-16.0	1.0	-10.6	14.2	-4.6	12.5	-6.2	5.4	30	0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	26.3	17.4	24.4	16.8	22.8	16.0	18.6	24.4	17.6	22.7	16.7	21.3	3.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.5	12.6	20.5	15.6	12.0	19.5	14.8	11.4	18.7	55.4	24.6	52.0	22.8	49.1	21.6	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.3	8.2	-15.0	30.0	4.1	1.5	-18.0	31.0	-20.4	31.9	-22.7	32.7	-25.6	33.8		
(5)			-15.5	20.6	4.1	0.8	-18.5	21.1	-20.9	21.6	-23.2	22.0	-26.2	22.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.8	-2.4	-2.5	1.9	5.7	10.9	13.6	16.2	16.1	12.2	7.8	1.8	-0.7		
	DBStd	7.81	4.70	4.63	4.12	4.03	4.09	3.92	3.66	3.54	3.46	3.71	3.69	3.66		
	HDD10.0	1858	386	349	252	139	40	9	1	1	14	89	246	332		
	HDD18.3	4266	644	582	509	379	229	149	87	87	184	328	496	590		
	CDD10.0	682	0	0	1	9	69	117	193	191	82	19	0	0		
	CDD18.3	48	0	0	0	0	1	7	21	18	2	0	0	0		
	CDH23.3	275	0	0	0	0	2	39	122	107	4	0	0	0		
	CDH26.7	37	0	0	0	0	0	3	18	16	0	0	0	0		
	WSAvg	3.7	3.7	4.0	4.0	4.1	4.1	3.6	3.4	3.2	3.5	3.7	3.6	3.2		
(15) Precipitation	PrecAvg	937	89	72	67	54	71	80	100	74	77	71	85	97		
	PrecMax	1183	205	189	160	113	178	136	151	196	184	232	152	242		
	PrecMin	685	3	11	13	5	13	25	38	25	16	18	4	30		
	PrecStd	138	42	40	38	25	40	32	31	36	34	44	37	45		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	10.4	15.3	18.9	23.5	27.2	28.9	28.7	23.8	19.2	11.3	9.1		
		MCWB	6.3	7.3	8.4	11.1	14.6	18.0	18.4	19.0	15.8	14.0	8.1	7.8		
	2%	DB	5.3	7.4	11.7	16.4	21.5	24.2	26.5	26.1	21.9	16.7	9.6	7.0		
		MCWB	4.8	5.2	7.5	10.2	13.5	16.5	17.6	17.4	15.4	12.7	7.9	6.0		
	5%	DB	4.2	5.3	9.2	14.3	19.6	22.3	24.5	24.1	19.7	14.4	8.3	5.7		
		MCWB	3.8	3.8	6.0	8.8	12.7	15.4	17.1	16.8	14.6	11.6	6.8	5.0		
	10%	DB	3.1	3.2	7.2	12.1	17.8	20.2	22.7	22.1	17.7	12.7	6.7	4.3		
		MCWB	2.7	2.2	5.2	7.7	11.9	14.5	16.3	16.1	13.8	10.5	5.6	3.8		
	0.4%	WB	6.7	7.8	10.0	12.0	15.9	19.0	20.0	20.1	17.7	14.8	9.6	8.1		
		MCDB	7.1	9.5	13.0	17.1	20.5	25.2	26.7	27.1	20.9	17.9	10.0	8.6		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	5.0	5.7	7.9	10.5	14.4	17.4	18.7	18.5	16.2	13.3	8.5	6.4		
		MCDB	5.2	7.0	11.2	15.3	19.4	22.6	24.6	24.2	19.8	15.6	9.4	6.8		
	5%	WB	3.9	3.9	6.6	9.2	13.4	16.3	17.7	17.4	15.2	11.9	7.1	5.1		
		MCDB	4.2	5.0	8.7	12.9	18.5	20.6	23.0	22.4	18.3	13.9	7.9	5.5		
	10%	WB	2.7	2.3	5.3	8.1	12.4	15.1	16.6	16.5	14.2	10.9	5.7	3.9		
		MCDB	3.1	3.1	7.0	11.6	17.0	19.1	21.6	20.9	17.1	12.5	6.4	4.2		
	5% DB	MDBR	3.7	4.6	5.5	7.5	8.0	7.6	8.2	8.0	6.9	5.7	4.0	3.4		
		MCDBR	4.9	6.5	8.9	10.8	10.7	10.4	10.8	10.9	9.6	7.9	6.1	4.8		
		MCWBR	4.7	4.9	5.5	6.0	5.7	5.4	5.4	5.3	5.0	4.8	4.9	4.2		
		MCWBR	4.6	5.8	7.8	9.7	9.5	9.4	10.1	9.5	8.5	7.1	5.5	4.2		
(40) Clear-Sky Solar Irradiance	taub		0.286	0.307	0.341	0.375	0.380	0.379	0.391	0.385	0.355	0.333	0.311	0.284		
		taud	2.444	2.406	2.374	2.320	2.330	2.346	2.310	2.351	2.418	2.481	2.489	2.447		
	Ebn at Noon		764	834	862	867	877	878	860	847	837	794	726	710		
		Edh at Noon	59	79	98	116	120	119	122	111	93	74	58	52		
(44) All-Sky Solar Radiation	RadAvg		0.79	1.53	2.60	4.09	4.86	5.32	5.04	4.35	3.13	1.84	0.87	0.59		
	RadStd		0.14	0.24	0.34	0.53	0.57	0.45	0.58	0.45	0.44	0.23	0.12	0.07		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)		N/A	N/A	N/A	N/A	N/A	+0.36	N/A	N/A	N/A	N/A

Nomenclature: See separate page